

Unit 3 – The Cellular Concept: Solved Answers

Subject: Mobile and Pervasive Computing (01CT0716)

4 MARKS QUESTIONS

Q1. Cellular system: 80 MHz bandwidth, 25 kHz simplex channels, 2 MHz control, cell area 12 km², total area 4320 km²

Given:

- Total bandwidth = 80 MHz
- Simplex channel width = 25 kHz → full-duplex channel width = $2 \times 25 = 50$ kHz
- Control channel spectrum = 2 MHz
- Cell area = 12 km², Total coverage area = 4320 km²

Step 1 – Total duplex channels Total channels = $80,000 \text{ kHz} \div 50 \text{ kHz} = \mathbf{1600 \text{ channels}}$

Step 2 – Control and voice channels

- Control channels = $2000 \text{ kHz} \div 50 \text{ kHz} = \mathbf{40 \text{ channels}}$
- Voice channels = $1600 - 40 = \mathbf{1560 \text{ channels}}$

Step 3 – Number of cells in the system Total cells = $4320 \text{ km}^2 \div 12 \text{ km}^2 = \mathbf{360 \text{ cells}}$

Step 4 – Channels per cell

- For N = 6: Voice/cell = $1560 \div 6 = \mathbf{260}$, Control/cell = $40 \div 6 \approx \mathbf{7}$
- For N = 3: Voice/cell = $1560 \div 3 = \mathbf{520}$, Control/cell = $40 \div 3 \approx \mathbf{13}$

Step 5 – System capacity ($C = S \times M$, where M = total cells / N)

- N = 6: $C = 1560 \times (360/6) = 1560 \times 60 = \mathbf{93,600 \text{ channels}}$
- N = 3: $C = 1560 \times (360/3) = 1560 \times 120 = \mathbf{187,200 \text{ channels}}$

Comment: Reducing cluster size from N = 6 to N = 3 **doubles** system capacity, because each cell gets more channels and the cluster repeats more often across the coverage area. However, a smaller N reduces the co-channel reuse ratio $Q = \sqrt{(3N)}$, bringing co-channel cells closer together and **increasing co-channel interference**. This illustrates the fundamental capacity-vs-interference trade-off in cellular design.

Q2. Handoffs for a mobile traveling at 72 km/h, cell radius 1.5 km, call duration 4 minutes

Given: $v = 72 \text{ km/h}$, $R = 1.5 \text{ km}$ (cell diameter $= 2R = 3 \text{ km}$), $T = 4 \text{ min} = 1/15 \text{ h}$

Distance traveled during the call: $d = v \times T = 72 \times (4/60) = 4.8 \text{ km}$

Assuming the mobile travels along a straight line through cell centers, time to cross one cell: $t_{\text{cell}} = 2R / v = 3/72 \text{ h} = 2.5 \text{ min}$

Number of handoffs expected: $N_{\text{handoff}} = \text{Call duration} / \text{Time per cell} = 4 / 2.5 = 1.6 \approx 2$ handoffs

The user is expected to experience **about 1-2 handoffs** during the call.

Q3. Micro-mobility vs Macro-mobility

Aspect	Micro-mobility	Macro-mobility
Scope	Movement within a small local domain (same access network/subnet)	Movement across different networks/domains (inter-domain)
Signaling overhead	Low — handled locally, doesn't involve the home network	High — requires signaling all the way to the home agent/HLR
Handoff performance	Fast, low latency (suited for frequent, local handoffs)	Slower, higher latency due to longer signaling path
Example protocols	Cellular IP, HAWAII, fast handover schemes	Mobile IP, GSM location update to HLR

Analysis: Micro-mobility protocols localize handoff signaling to reduce delay and control traffic, making them ideal for users moving frequently within a small area (e.g., between cells of the same city). Macro-mobility protocols must update the home network database, incurring larger delays — acceptable since such moves (e.g., between cities/countries) are less frequent.

Q4. Handoff management for a high-speed train scenario

High-speed trains cross many small cells rapidly, so:

- **Frequent handoffs** would occur if normal-sized cells were used, overloading the MSC.
- The **umbrella cell approach** is most suitable: a large "umbrella" cell (using high-power, high-antenna transmitters) covers the entire route for high-speed users, while

small microcells beneath serve slow-moving/pedestrian users.

- This way, the train stays connected to one large cell for a long duration, drastically **reducing the number of handoffs** and MSC signaling load.
- Alternatively, **soft handoff** (as in CDMA) or **network-assisted, predictive handoff** (as in LTE/5G) can be used, where the network anticipates the next cell based on trajectory and speed.

Conclusion: The umbrella cell (or a dedicated high-power "fast lane" cell along the track) is the most practical solution for high-speed train handoff management.

Q5. Coordination of link layer and network layer mobility

- **Link layer (Layer 2) mobility:** Handles handoff at the radio level — switching the radio connection from one base station/access point to another (e.g., changing BTS/BSC in GSM, or AP in Wi-Fi). It is fast and deals with signal strength, channel allocation.
- **Network layer (Layer 3) mobility:** Handles maintaining IP-layer connectivity as the device changes its point of attachment (e.g., Mobile IP updating the care-of address, GSM registering with a new VLR).

Coordination: For seamless handoff, Layer 2 first re-establishes the radio link quickly (low latency), and Layer 3 then updates routing/location information so packets are correctly delivered to the new point of attachment. This two-layer coordination minimizes both **radio-level interruption** and **packet loss/misrouting**.

Application to macro/micro mobility:

- **Micro-mobility** relies mainly on **Layer 2** mechanisms (fast, local re-routing without informing the home network).
 - **Macro-mobility** relies on **Layer 3** mechanisms (Mobile IP-style home agent updates) since the device moves to an entirely different network domain.
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Q6. Cellular system: 60 MHz bandwidth, 50 kHz simplex channels, 1 MHz control, cell 9 km², total area 3630 km², N reduced to 4

Given: BW = 60 MHz, simplex = 50 kHz → duplex channel = 100 kHz, control = 1 MHz, cell area = 9 km², total area = 3630 km²

Step 1 - Total duplex channels Total = 60,000 kHz ÷ 100 kHz = **600 channels**

Step 2 - Control and voice channels

- Control = 1000 ÷ 100 = **10 channels**
- Voice = 600 – 10 = **590 channels**

Step 3 - Number of cells Total cells = $3630 \div 9 \approx 403$ cells

Step 4 - Channels per cell for N = 4

- Voice/cell = $590 \div 4 = 147.5 \approx 147$
- Control/cell = $10 \div 4 = 2.5 \approx 2-3$

Step 5 - System capacity $C = S \times M = 590 \times (403/4) \approx 590 \times 100.75 \approx 59,440$ channels

Comment: Reducing the cluster size to $N = 4$ increases the number of channels available per cell and overall system capacity, but the co-channel reuse ratio $Q = \sqrt{(3 \times 4)} = \sqrt{12} \approx 3.46$ is smaller, meaning co-channel cells are relatively closer, raising the risk of co-channel interference. Careful power control/sectoring would be needed to maintain acceptable signal quality.

Q7. Prove $Q = \sqrt{(3N)}$, where $N = i^2 + ij + j^2$

Setup: In a hexagonal cellular layout, adjacent cell centers are separated by a distance of $\sqrt{3}R$ (where R is the cell radius), and the lattice directions are separated by an angle of 60° .

To find a co-channel cell, we move i cells along one hexagonal chain, turn 60° , then move j cells. In vector terms, if $\mathbf{u}$ and $\mathbf{v}$ are the two lattice basis vectors (each of magnitude $\sqrt{3}R$, separated by 60°), the co-channel distance vector is:

$$\mathbf{D} = i \cdot \mathbf{u} + j \cdot \mathbf{v}$$

Using the law of cosines (angle between $\mathbf{u}$ and $\mathbf{v}$ directions = 120° , since consecutive hex-grid axes are offset by 60° but vector addition uses the supplementary angle):

$$D^2 = (i\sqrt{3}R)^2 + (j\sqrt{3}R)^2 - 2(i\sqrt{3}R)(j\sqrt{3}R)\cos(120^\circ)$$

Since $\cos(120^\circ) = -1/2$:

$$D^2 = 3R^2i^2 + 3R^2j^2 + 3R^2ij \quad D^2 = 3R^2(i^2 + ij + j^2) \quad D^2 = 3R^2N$$

Therefore:

$$D = R\sqrt{(3N)}$$

Dividing both sides by R :

$$Q = D/R = \sqrt{(3N)} \blacksquare$$

This confirms that the co-channel reuse ratio Q depends only on the cluster size N , and increases as N increases (larger clusters give better isolation between co-channel cells, at the cost of lower capacity).

6 MARKS QUESTIONS

Q1. How does LTE provide stable video calls at 150 km/h (e.g., on a train)?

LTE handles high-speed mobility through several mechanisms:

1. **Hard handoff with network-assisted measurement:** The UE (User Equipment) continuously reports signal strength (RSRP/RSRQ) of serving and neighboring cells to the eNodeB, enabling fast, "break-before-make" handoff decisions.
2. **X2 interface handover:** Direct signaling link between neighboring eNodeBs allows handover preparation and execution with minimal core-network involvement, reducing handover latency.
3. **Predictive/early handover triggering:** Hysteresis margins and Time-to-Trigger (TTT) parameters are tuned for high-speed scenarios to initiate handover earlier, before signal quality drops too low.
4. **Larger/umbrella cells along high-speed corridors:** Reduces handover frequency for fast-moving users (similar to the umbrella cell concept).
5. **Robust link adaptation (AMC):** Adaptive Modulation and Coding adjusts the modulation scheme in real time to compensate for rapidly changing channel conditions caused by Doppler shift at high speed, maintaining link quality during video streaming.
6. **Buffering/QoS bearers:** Dedicated bearers with guaranteed bit rate (GBR) for video ensure packet loss during brief handover gaps doesn't disrupt playback.

Together, these mechanisms ensure the video call session is maintained with minimal interruption despite frequent cell crossings at high speed.

Q2. Mobility and handoff management in 5G for uninterrupted video streaming on a high-speed train

5G improves on LTE with:

1. **Dual connectivity:** The device can maintain simultaneous connections to two cells (e.g., a low-frequency macro cell for mobility/coverage and a high-frequency small cell for capacity), so losing one connection during handover doesn't interrupt the session.
2. **Make-before-break handover:** Unlike LTE's typical hard handover, 5G supports establishing the new connection before releasing the old one, virtually eliminating the interruption gap.
3. **Beam management:** 5G uses beamforming with rapid beam-switching to track a fast-moving user without a full cell handover.
4. **Network slicing and edge computing (MEC):** Ensures video content is cached/processed close to the user, reducing the impact of any handover delay on perceived video quality.

5. **AI/ML-based mobility prediction:** 5G networks can predict a train's trajectory (fixed route) and pre-configure handovers, further reducing latency.
6. **Massive MIMO and higher capacity:** Ensures enough bandwidth remains available even as the user moves through cells with varying loads.

Result: These mechanisms combine to give the train passenger a smooth, high-quality, uninterrupted video streaming experience despite constant high-speed cell transitions.

Q3. Applying frequency reuse to design a cellular layout for a city with limited spectrum and high user density

Design approach:

1. **Choose a small cluster size (N)** such as $N = 3, 4$, or 7 to maximize the number of times the limited spectrum S is reused across the city (since capacity $C = S \times (\text{Total Area}/\text{Cell Area})/N$ — smaller N gives higher capacity).
2. **Reduce cell radius (cell splitting):** In high-density zones (e.g., business districts), split cells into smaller ones so the same channel set serves more users through additional reuse.
3. **Use sectoring (e.g., 120° sectors):** Improves the signal-to-interference ratio (SIR), allowing the network to safely use a smaller N (higher capacity) without excessive co-channel interference.
4. **Employ the microcell zone concept** along dense traffic corridors (e.g., main roads), reducing handoffs and improving coverage without adding new base stations.
5. **Apply power control:** Keep transmit power to the minimum necessary, reducing co-channel interference and enabling tighter frequency reuse.
6. **Check the co-channel reuse ratio:** $Q = \sqrt{3N}$ must be verified against the required SIR for acceptable call quality — a trade-off between N (capacity) and Q (quality).

Conclusion: For a high-density city with limited spectrum, the optimal design typically combines a small cluster size ($N = 3$ or 4), cell splitting in congested areas, and 120° sectoring, giving maximum capacity while keeping interference within acceptable limits.

8 MARKS QUESTIONS

Q1. Principle of cell splitting; prove 12 dB power reduction needed for path loss exponent $n = 4$

Principle of Cell Splitting: Cell splitting subdivides a congested cell into smaller cells, each with its own base station, reduced antenna height, and reduced transmit power. This increases the number of times channels are reused within the same geographic area, thereby increasing system capacity **without requiring additional spectrum**. New cells

with radius $R/2$ are created within the original cell of radius R (see Chapter 3, Figures 3.8–3.9).

Proof of power reduction:

Average received power at distance d from a transmitter is modeled as:

$$P_r(d) = P_t \cdot (d_0/d)^n \text{ (where } n \text{ is the path loss exponent)}$$

For a cell of radius R with transmit power P_t , the received power at the cell boundary is:

$$P_r(R) = P_t \cdot R^{-n} \text{ (proportional relation)}$$

When the cell is split so the new radius is $R/2$, to maintain the **same signal quality at the new cell boundary** (same received power level), the new transmit power $P_{t'}$ must satisfy:

$$P_t \cdot R^{-n} = P_{t'} \cdot (R/2)^{-n}$$

Solving for $P_{t'}$:

$$P_{t'} = P_t \cdot (R/2)^n / R^n = P_t \cdot (1/2)^n$$

For $n = 4$:

$$P_{t'} = P_t \times (1/2)^4 = P_t / 16$$

Converting to dB:

$$\text{Reduction (dB)} = 10 \log_{10}(16) = 10 \times 1.204 = \mathbf{12.04 \text{ dB} \approx 12 \text{ dB}}$$

Conclusion: When cell radius is halved during cell splitting, and the path loss exponent is $n = 4$, the transmit power of the new (smaller) cell's base station must be reduced by approximately **12 dB** (i.e., to 1/16th of the original power) to maintain the same signal-to-interference quality at the new, smaller cell boundary. ■

Q2. Role of VLR and HLR in supporting mobility in GSM networks

- **HLR (Home Location Register):** A permanent, centralized database maintained by the subscriber's home network operator. It stores permanent subscriber information — IMSI, subscribed services, and (importantly) a pointer to the **current VLR** where the subscriber is registered.
- **VLR (Visitor Location Register):** A temporary database associated with each MSC, storing information about all subscribers **currently located** in that MSC's service area, including a copy of subscriber data fetched from the HLR.

Interaction during location update:

1. When a mobile moves into a new MSC/VLR area, it sends a **Location Update Request** to the new VLR.
2. The new VLR contacts the subscriber's **HLR** (using the IMSI) to verify and retrieve subscriber profile data.

3. The HLR updates its record to point to the **new VLR** as the subscriber's current location, and may instruct the **old VLR to delete** the subscriber's temporary record.
4. The new VLR stores the subscriber data locally so that future call setups within that area can be processed quickly, without repeatedly querying the HLR.

During an incoming call: The call is first routed to the subscriber's HLR (via the subscriber's number), which queries the current VLR for the subscriber's exact location (via the MSC), enabling the call to be routed to the correct BTS/cell.

This two-tier database structure (permanent HLR + temporary VLR) allows GSM to efficiently track a highly mobile subscriber base while minimizing signaling overhead for local operations like call setup.

Q3. Role of frequency reuse in wireless network systems

Concept: Frequency reuse is the technique of allocating the same set of radio channels to different cells that are physically separated far enough apart to keep co-channel interference at an acceptable level. Given S total duplex channels divided among N cells in a cluster ($S = kN$), the cluster pattern is repeated M times across the coverage area, giving total system capacity $C = MS = MkN$.

How it enhances spectrum efficiency:

- Without frequency reuse, a single high-power transmitter serving a large area could only offer a limited number of channels (S) to the whole region.
- With frequency reuse, the *same* S channels are reused in every cluster repetition, so a system with M cluster repetitions supports **M times more simultaneous users** than a single-cell system — using the **same fixed spectrum**.

Diagram concept (as in the chapter, e.g., 7-cell reuse): A cluster of $N = 7$ cells (labeled A–G) each use a different channel group. This 7-cell cluster pattern is tiled repeatedly across the city. Cells labeled "A" in different clusters use the same channel group, but they are separated by a reuse distance D , calculated from $D = R\sqrt{3N}$, keeping co-channel interference low despite reusing the same frequencies.

Support for large-scale coverage: Because frequency reuse allows unlimited geographic scaling (simply tile more clusters) without needing more spectrum, it enables large-scale, city/region-wide coverage using a small allocated bandwidth — this is the foundational principle that makes modern cellular networks scalable.

Q4. Handoff strategies: Hard handoff vs. Soft handoff

Hard Handoff ("break-before-make"):

- The connection to the old base station is **released before** the connection to the new base station is established.
- Used in FDMA/TDMA systems like GSM.
- Simpler to implement, requires only one channel at a time.
- Risk: brief signal interruption; if the new channel isn't acquired properly, the call may drop.
- **Scenario:** A mobile moving from BS1 to BS2 at a clear cell boundary in GSM — the switch happens at a defined signal threshold crossing.

Soft Handoff ("make-before-break"):

- The mobile **simultaneously connects** to both the old and new base stations for a period, combining signals (e.g., via RAKE receiver in CDMA), before dropping the old connection.
- Used in CDMA systems, where all cells can share the same frequency.
- Provides smoother transition, no perceptible interruption, improved reliability at cell edges.
- Requires more network resources (channels at two base stations simultaneously).
- **Scenario:** A CDMA mobile at the boundary between two cells maintains links to both BS1 and BS2 simultaneously, selecting/combining the stronger signal, until it moves clearly into the new cell's coverage.

Comparison Summary:

Feature	Hard Handoff	Soft Handoff
Connection overlap	None	Yes (temporary)
Systems	GSM, TDMA/FDMA	CDMA
Call drop risk	Higher	Lower
Resource usage	Lower	Higher
User experience	Brief click/gap possible	Seamless

Q5. Mobility management in GSM: Location Update vs. Handover

Location Update:

- Occurs when a mobile moves between **location areas** (groups of cells) while **idle** (not in a call).

- The mobile informs the network (via the new VLR/MSC) of its new location so incoming calls/SMS can be routed correctly.
- Involves updating the **HLR** and **VLR** databases (as described in Q2 above).
- Relatively infrequent and can tolerate more signaling delay since no active call is in progress.

Handover (Handoff):

- Occurs when a mobile moves between cells **during an active call**.
- Requires **real-time**, low-latency processing to switch the radio channel from the old BTS to the new one without interrupting the ongoing conversation.
- Managed by the MSC/BSC using signal strength measurements, without needing to update the HLR (unless the mobile also crosses into a new location area).

Key differences:

Aspect	Location Update	Handover
Call state	Mobile is idle	Mobile is in an active call
Urgency	Can tolerate delay	Must be near-instantaneous
Databases involved	HLR, VLR	BSC, MSC (mainly)
Trigger	Crossing location area boundary	Signal quality degradation at cell boundary

Q6. Why is a hexagonal cell shape preferred over square or triangular shapes?

Geometric shapes that can tile a plane without overlap or gaps (for equal-area coverage) are limited to squares, equilateral triangles, and regular hexagons. Comparing these three:

1. **Fewest cells to cover a region:** For a given cell radius R (maximum distance from center to edge), the hexagon has the **largest area** among the three shapes. This means fewer hexagonal cells are needed to cover a given geographic region compared to squares or triangles, reducing the number of base stations required.
2. **Closer approximation to a circle:** A hexagon most closely approximates the circular radiation pattern of an omnidirectional antenna, minimizing overlap and gaps between adjacent cells compared to squares or triangles.
3. **Uniform distance to neighbors:** In a hexagonal grid, all six neighboring cells are equidistant from the center cell, simplifying frequency planning and providing consistent co-channel interference characteristics — squares and triangles do not offer this uniform symmetry (some neighbors are closer via edges vs. corners).

4. **Efficient tessellation:** Hexagons tessellate perfectly (no wasted overlap area) while also minimizing the total cell boundary length needed to cover an area.

Conclusion: The hexagon offers the best trade-off of coverage efficiency (largest area per cell radius), symmetric interference geometry, and closeness to real antenna radiation patterns — which is why it is the standard model used in cellular network design, even though real coverage areas are irregular in practice.

Q7. Macro mobility vs. Micro mobility (with real-life examples in Mobile IP systems)

Macro Mobility:

- Refers to a mobile device moving across **different administrative domains or networks** (e.g., between different ISPs, different cities, or countries).
- Managed at the **network layer**, typically using protocols like **Mobile IP**, where the mobile node registers a new **Care-of-Address (CoA)** with its **Home Agent** whenever it changes networks.
- Involves higher signaling overhead and latency since the Home Agent (potentially far away) must be updated.
- **Example:** A user traveling from Mumbai to a different city (or country) with their phone connected to Wi-Fi, changing from one ISP's network to a completely different one — Mobile IP routes traffic through the Home Agent to the new network's Foreign Agent/CoA.

Micro Mobility:

- Refers to movement **within** a single administrative domain or a small local network (e.g., between cells/access points of the same operator/campus network).
- Managed more locally, without needing to inform the distant Home Agent every time — protocols like **Cellular IP** or **HAWAII** handle handoffs locally within the domain.
- Lower latency and signaling overhead, suited to frequent, short-distance moves.
- **Example:** A user walking across a university campus, moving between different Wi-Fi access points that belong to the same campus network — the local network handles the handoff without contacting an external Home Agent.

Summary Table:

Feature	Macro Mobility	Micro Mobility
Domain	Across different networks	Within same network/domain
Protocol	Mobile IP	Cellular IP, HAWAII, local handoff schemes
Latency	Higher	Lower
Signaling overhead	High (Home Agent contacted)	Low (handled locally)
Example	Switching operators/cities/countries	Moving between cells of the same operator/campus

End of Unit 3 Answers